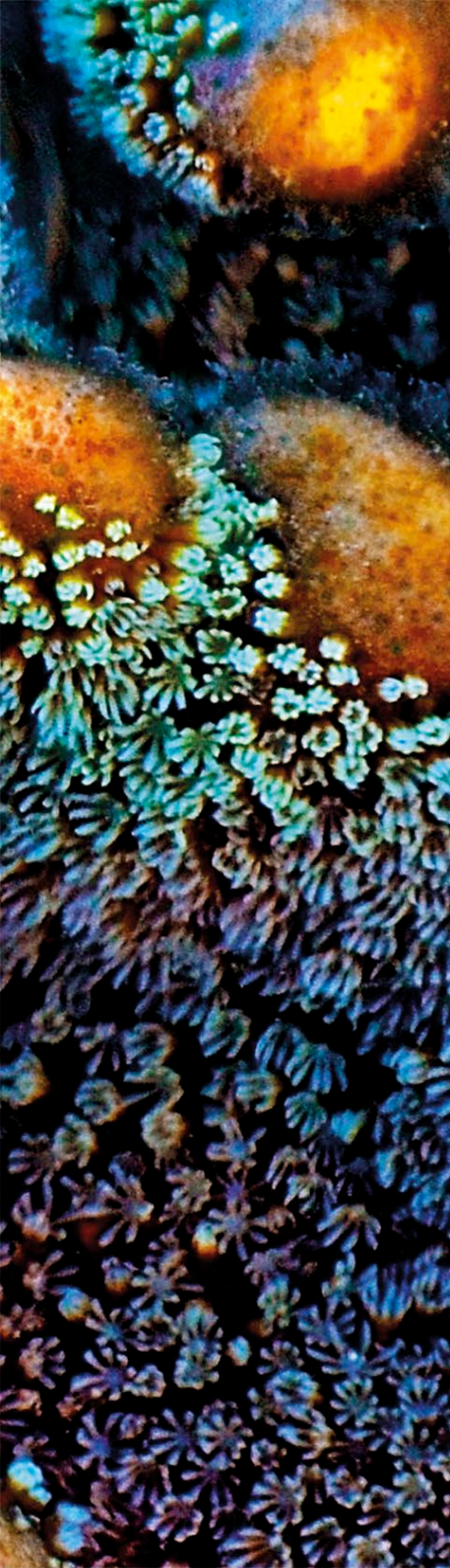




The Great Barrier Reef

The world's most extensive coral reef system, often described as the largest structure ever built by living organisms

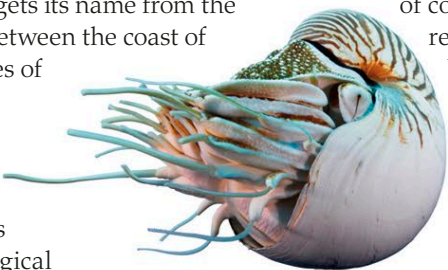


The Great Barrier Reef stretches for 1,400 miles (2,300km) over a part of the Coral Sea, which lies off the coast of Queensland, Australia. Rather than a single unbroken reef, it is composed of more than 2,900 individual small reefs of various types and 900 coral islands, covering a total area of some 133,000 square miles (344,400 square km). The whole structure has been built through the activity of animals called coral polyps (see panel, below). The reef gets its name from the fact that it acts as a barrier between the coast of Australia and the large waves of the Pacific Ocean.

Biological diversity
Teeming with marine life, the Great Barrier Reef boasts an astonishingly high biological

diversity, including about 350 species of stony corals and several hundred species of soft corals, as well as 1,500 species of fish, 17 species of sea snakes, 30 species of whales, dolphins, and porpoises, and 6 species of sea turtles. Over 200 bird species visit the reef, or nest or roost on the islands.

Unfortunately, in recent years the Great Barrier Reef has become severely threatened, the biggest cause of concern being rising ocean temperature as a result of global warming. This leads to coral bleaching (see p.286), a process that can ultimately kill corals permanently.



< LIVING FOSSIL
The chambered nautilus is one of over 4,000 species of mollusks that inhabit the reef. It has evolved little over many millions of years.

The Great Barrier Reef is the **size of** about **48 million international football pitches**



△ BUDDING LIFE
A coral forms when a polyp attaches itself to a rock underwater. It then divides to form a colony—this one contains hundreds of polyps.

< STRUCTURAL COMPLEXITY
A variety of forms exist in even a small reef section, with patches and platforms of limestone, as well as ribbonlike structures.

STONY AND SOFT CORALS

Corals are made up of many individual animals called polyps. There are two main types of corals, hexacorals (stony corals) and octacorals (soft corals). A significant difference is that it is mainly hexacoral polyps that secrete calcium carbonate, making them the principal reef builders.

